

Viscosity of a Non-Newtonian Fluids

Final Lab Report
Unit Operations Lab 1
November 11, 2025

Group 3
Adil Feratovic
Hussein Rezk
Anna Cai
Aaron Casas

1. Executive Summary (Examples at end of the Guideline) (10 pts)

This experiment utilized a capillary tube viscometer to investigate the behavior of a 1 wt% solution of Carbopol 940. The viscosity parameters of m and n were found for the solution using capillary tubes of varying lengths and diameters at different pressures. These parameters are important in predicting fluid behavior when designing chemical engineering processes.

The experiment was conducted by pressurizing the solution to 50, 60, and 70 psig and measuring the flow rate through capillary tubes with diameters of 0.062 and 0.07 inches and lengths of 7.5, 15.3, and 23 cm. Plots of shear stress vs velocity gradient at different diameters and lengths showed pseudoplastic behavior along with the correlation to the velocity gradient. The viscosity parameters were then calculated with the Ostwald-de Waele model.

The results show that the viscosity parameter m varied with tube length and diameter. Comparing m values of one diameter across the three lengths finds an average m of 16.05 and 26.36 Pa*s for diameters of 0.062 and 0.07 in, respectively. The parameter of n was also determined and compared similarly, with an average of 0.539 and 0.453 for diameters of 0.062 and 0.07 in, respectively. As the average n value was found to be less than 1, it can be concluded that the Carbopol 940 solution behaves as a pseudoplastic fluid with shear-thinning properties.

2. Introduction (10 pts)

Viscosity is a fundamental fluid property that quantifies a fluid's resistance to deformation under shear stress. In many industries, such as cosmetics, pharmaceuticals, and the food and beverage sector, precise control of viscosity is crucial for achieving the desired product quality and manufacturing consistency. Unlike Newtonian fluids, which maintain constant viscosity regardless of shear rate, non-Newtonian fluids display variable viscosity depending on the applied shear. Their complex behavior requires models, such as the Ostwald-de Waele power law, to accurately describe their flow characteristics. Understanding the behavior of non-Newtonian fluids is essential for designing efficient production processes. For example, in chemical reactors, fluid viscosity plays a critical role in determining the efficiency of mixing, heat transfer, and reaction rates. In the food industry, understanding and controlling the behavior of products such as sauces or spreads affects consistency, texture, and mouthfeel. Likewise, in pharmaceutical manufacturing, predictable flow behavior of gels during production ensures uniform dosing, consistent formulation, and reliable application performance.

Carbopol 940, a cross-linked polyacrylic acid polymer, is a widely used non-Newtonian fluid that exhibits a shear-thinning behavior. This property enables materials, such as cosmetic creams and pharmaceutical gels, to flow easily under stress while maintaining their thickness at rest, ensuring both ease of application and structural stability. Understanding how Carbopol 940's viscosity responds to different shear conditions is critical for optimizing the process design and product performance.

In this experiment, the consistency index and flow behavior index were determined using the Ostwald-de Waele model under varying pressures (50, 60, and 70 psig) and capillary tube dimensions (diameters of 0.062 and 0.07 inches, and lengths of 7.5, 15.3, and 23 cm.). The setup of the experiment consisted of a pressurization vessel connected to interchangeable capillary tubes, allowing for precise control of flow rate and pressure. Measured flow rates were used to calculate shear stress and shear rate, from which the viscosity parameters were derived.

These results provide valuable insights into Carbopol 940's non-Newtonian flow characteristics, supporting the development of efficient processing systems for materials where rheological behavior influences mixing, heat transfer, coating, and dispensing operations. Understanding these parameters helps improve product quality and process control across a wide range of applications.

3. Theory (10 pts)

The fluid property known as viscosity is a measure of a liquid's thickness. Viscosity is also known as the ratio of shear stress and shear rate. Equation 1 describes the relationship between viscosity, velocity gradient, and shear stress rate when a liquid behaves as a Newtonian fluid.

$$(1) \tau = -\mu (\nabla v + \nabla v^T + \frac{2}{3}(\mu - k)(\nabla \cdot v)\delta)$$

The second term in the equation can be neglected as the fluid in this experiment was very thick and viscous, therefore not very compressible. Equation 1 ultimately simplifies to Equation 2.

$$(2) \tau = -\mu (\nabla v + \nabla v^T)$$

Equation 2 then simplifies to Equation 3, as the fluid flow occurs in one direction.

$$(3) \tau_{xy} = -\mu \left(\frac{dv_x}{dy} \right)$$

For flow going through a pipe, we focus on the next equation.

$$(4) \tau_{rz} = -\mu \left[\frac{\partial v_r}{\partial z} + \frac{\partial v_z}{\partial r} \right]$$

The reason the focus is on equation 4 is that, in the experiment, a thick, viscous fluid flows through a cylindrical pipe, which describes how viscosity is affected by stress and velocity per unit length.

Since the experiment involves a non-Newtonian fluid, we use a different equation to model the relationship between stress, velocity, and viscosity. The following equation is used,

$$(5) \tau = \pm m \dot{\gamma}^n$$

Where m and n are the empirical constants acquired from graphing $\log\left(\frac{\Delta PR}{2L}\right)$ vs $\log\left(\frac{Q}{\pi R^3}\right)$

Equation 5 is also written in the following form

$$(6) \tau_{xy} = -m \dot{\gamma}^{n-1} \frac{dv_x}{dy} = -\mu_{app} \frac{dv_x}{dy}$$

Where μ_{app} is the apparent viscosity, which depends on the rate of shear experienced on a fluid. The relationship between the velocity gradient in the fluid and the shear stress is given in terms of the two constants m and n . If n is greater than 1, the fluid behavior is that of a dilatant fluid; if n is less than 1, we have a pseudo-plastic fluid, and if $n=1$, the liquid behaves in a Newtonian manner.

For cylindrical coordinates,

$$(7) \tau_{rz} = \pm m \dot{\gamma}^n$$

Then, the Hagen–Poiseuille equation relates the volumetric flow rate, the pressure drop across the capillary, the dimensions of the capillary, and the viscosity. In this experiment, an analytical equation is used for the flow rate of a non-Newtonian fluid, as well as an equation that follows the Ostwald-de Waele viscosity model, which is labeled as Equation 8.

$$(8) m \dot{\gamma}^n$$

Where m and n are the parameters of the Ostwald-deWaele model. The parameter n is obtained from plotting $\log\left(\frac{\Delta PR}{2L}\right)$ vs $\log\left(\frac{Q}{\pi R^3}\right)$ and taking its slope. To obtain m , the intercept value of equation 9 is used to solve for m .

$$(9) \log \{ m \dot{\gamma}^n \}$$

Additionally, each fluid has a different behavior determined by the shear stress, and the velocity gradient can be represented by the plot provided in the “Non-Newtonian Viscosity Lab Manual.” [2]

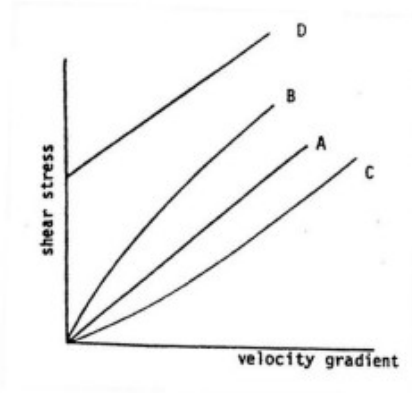


Figure 2. Relationship between strain and shear for different fluids: (A) Newtonian fluid; (B) psuedo-plastic fluid; (C) dilatant fluid; (D) Bingham fluid.

Figure 1: Different fluid behavior by type in Shear Stress V Velocity Gradient Plot

4. Results (15 pts)

Determining m and n parameters

The following plots were obtained by plotting $\log\left(\frac{\Delta PR}{2L}\right)$ vs $\log\left(\frac{Q}{\pi R^3}\right)$ at a total of 6 different capillary tubes at 3 different pressures. [2] All the graphs show very similar trends, as the curves have larger x and y values due to larger pressure drops, where the pressure drop is directly related to $\log\left(\frac{\Delta PR}{2L}\right)$ in the y-axis and related to volumetric flow rate in the x-axis.

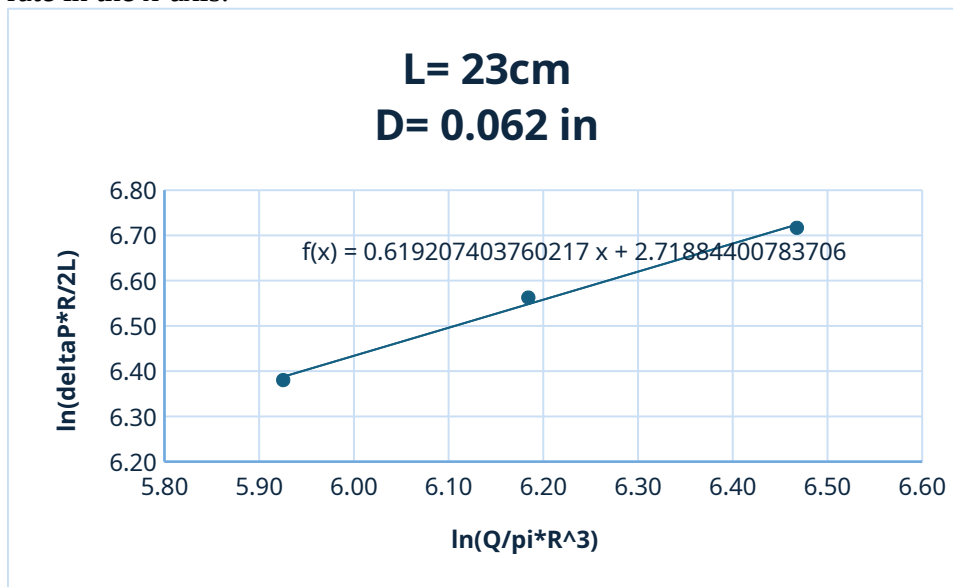


Figure 2: Relationship of pressure gradient and volumetric flow rate with length of 23 cm and diameter of 0.062 inches

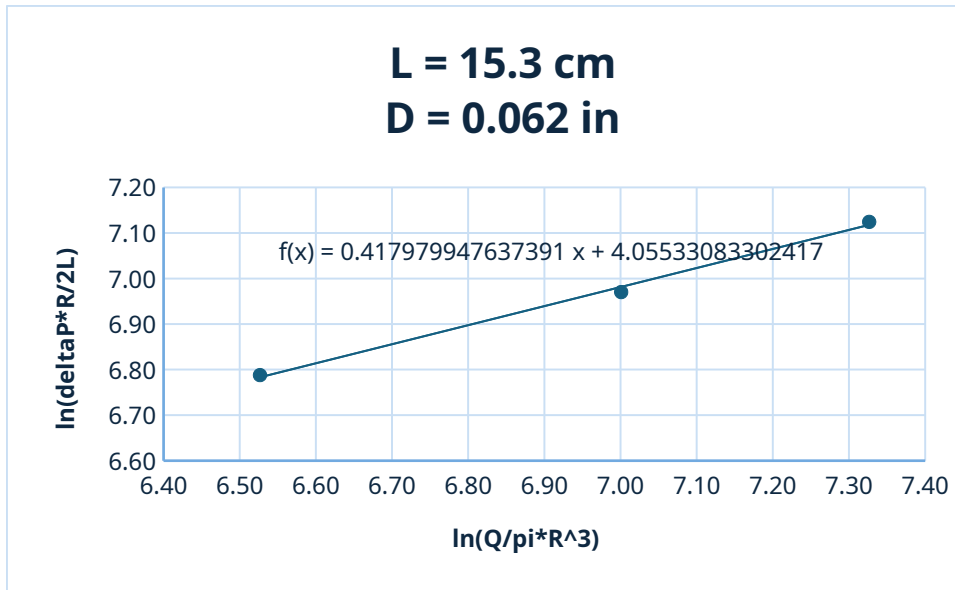


Figure 3: Relationship of pressure gradient and volumetric flow rate with length 15.3cm and diameter 0.062 inches

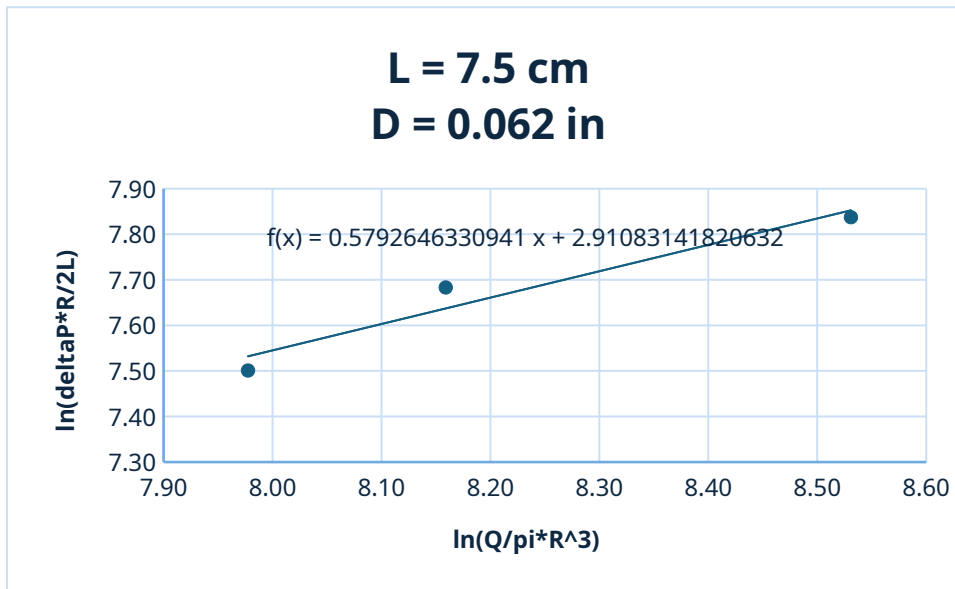


Figure 4: Relationship of pressure gradient and volumetric flow rate with length 7.5cm and diameter 0.062 inches

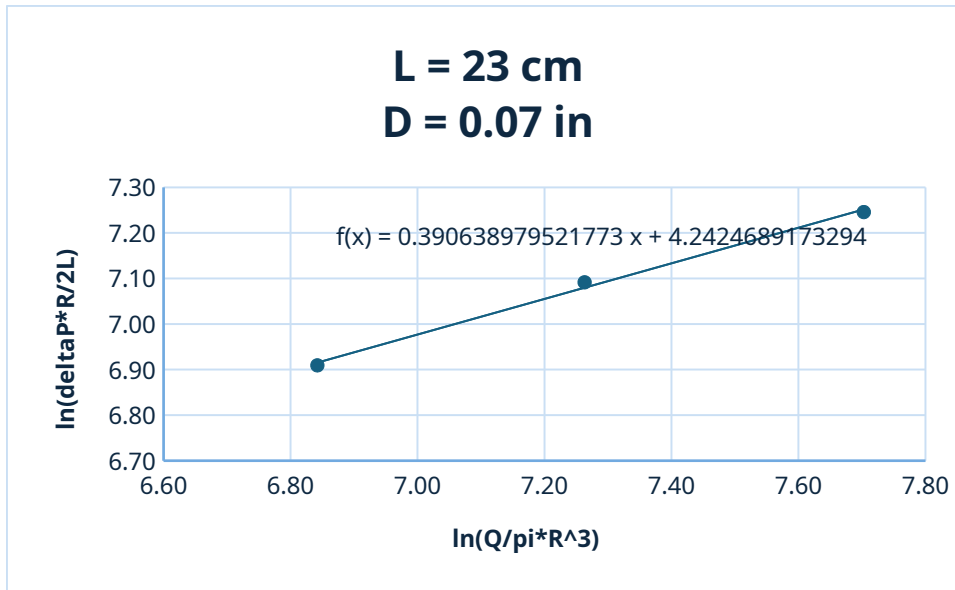


Figure 5: Relationship of pressure gradient and volumetric flow rate with length 23 cm and diameter 0.07 inches

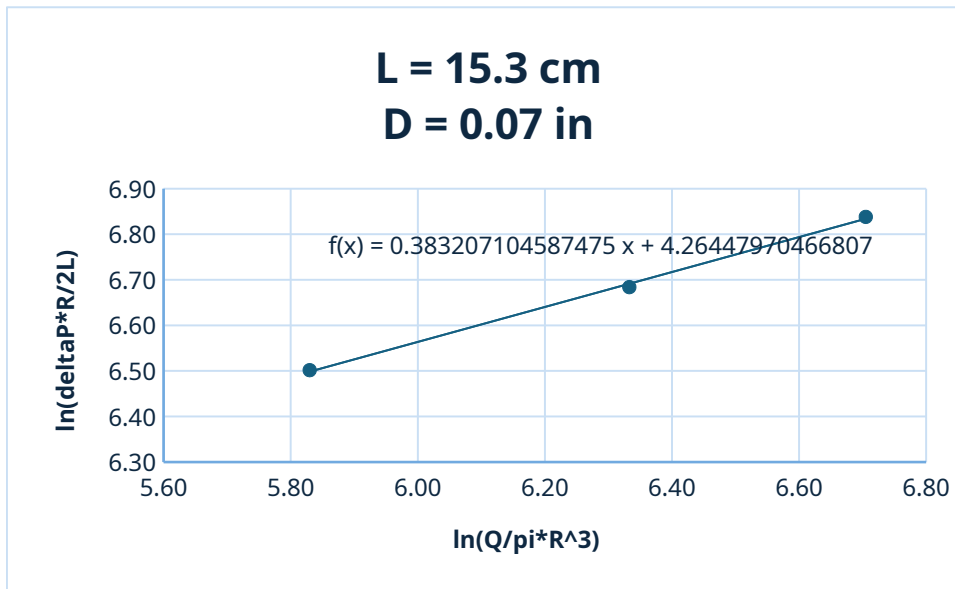


Figure 6: Relationship of pressure gradient and volumetric flow rate with length 15.3 cm and diameter 0.07 inches

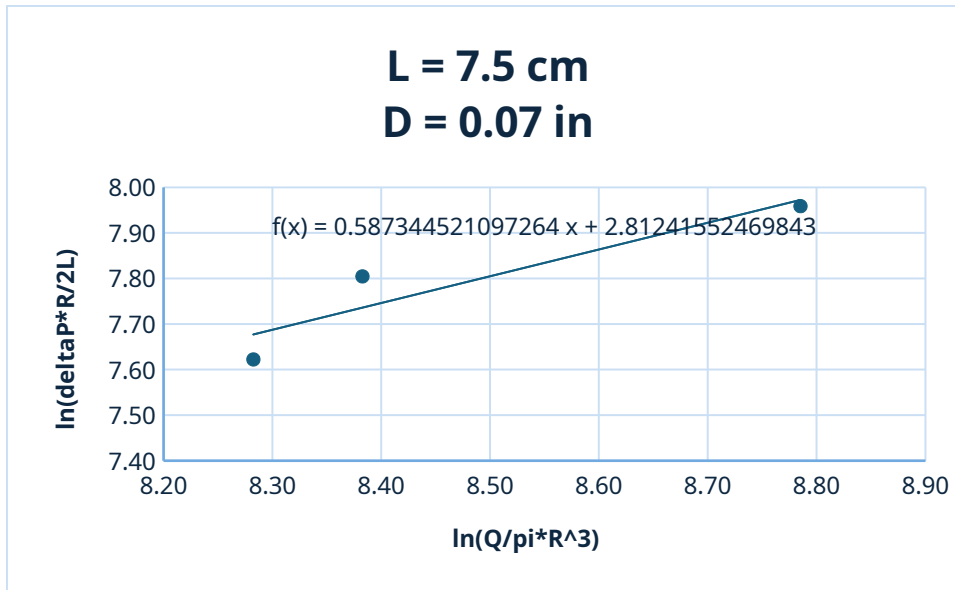


Figure 7: Relationship of pressure gradient and volumetric flow rate with length 7.5 cm and diameter 0.07 inches

m	Diameter	
	(in)	
length (cm)	0.062	0.07
23	12.15	35.60
15.3	28.53	36.77
7.5	7.471	6.707
avg	16.05	26.36

Table 1: *M* parameter corresponding to Diameter and length

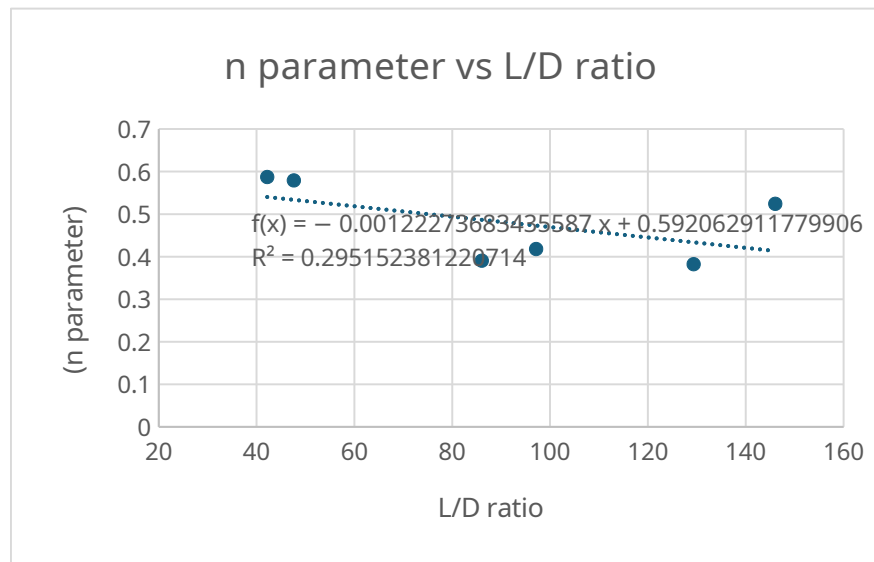
Looking at Table 1, the *m* parameter is the largest at the largest length and at the shortest diameter, while the smallest value occurs at the second largest diameter and length

To determine the *m* parameter, equation (9) was used, and the equation was set equal to the corresponding y-intercept.

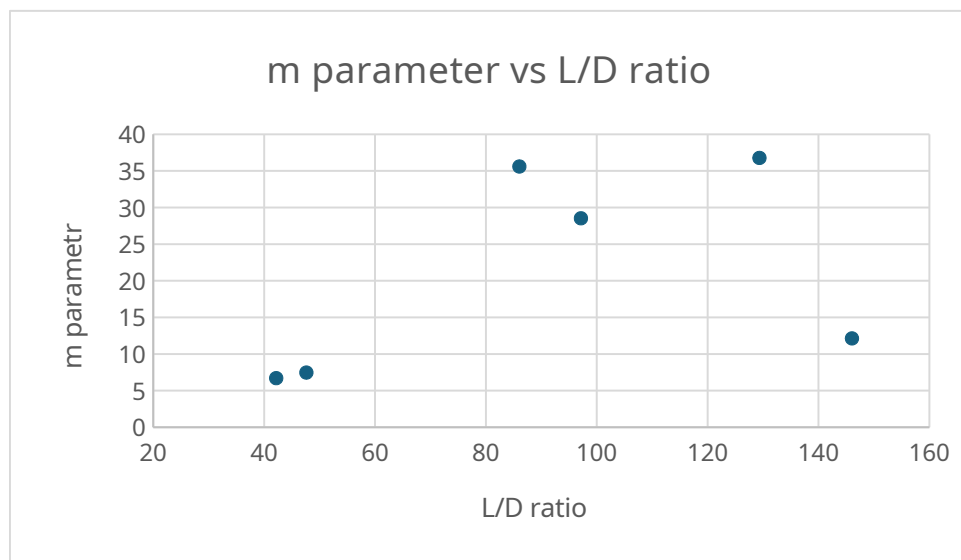
n	Diameter(in)	
length (cm)	0.062	0.07
23		0.390
	0.6192	6
15.3		0.382
	0.418	3
7.5		0.587
	0.5793	3
avg	0.5388	0.453

Table 2: n parameter corresponding to Diameter and length

Looking at Table 2, we see that the largest n parameter corresponds to a midsize length and the largest diameter. Therefore, it can be concluded that if the m parameter has a small value, its corresponding n parameter will be larger. Additionally, the value of n corresponds to the slope of its respective line.

Figure 8: Relationship between n parameter and L/D ratio

Looking at Figure 8, there is a similar trend from Table 2, where the n parameter does not have a linear relationship with the length and diameter of the capillary. At the smaller L/D ratio, the n parameter is at its largest, and in the midsize L/D ratio, the n parameter increases. Still, it later increases at the smallest length and the largest diameter.

Figure 9: Relationship between m parameter and L/D ratio

Looking at Figure 9, it further reinforces that the m and n parameters don't necessarily have a linear relationship with the length and diameter of the capillary. It also reinforces the relationship that lower length and diameter typically result in a high n parameter and a low m parameter, thereby concluding that the m and n parameters have an inverse relationship.

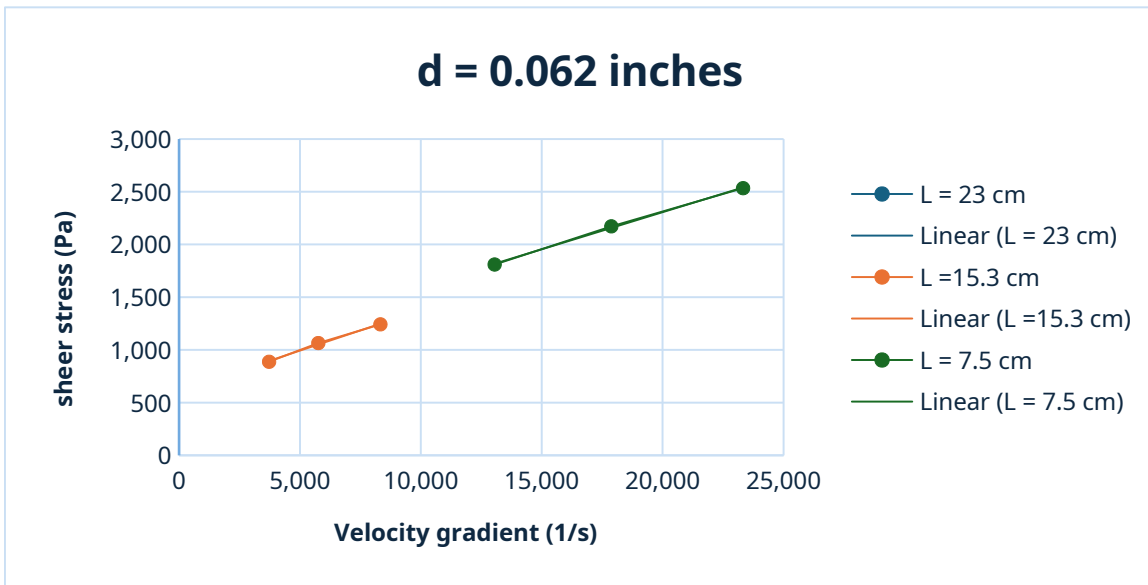


Figure 10: All ratios for diameter 0.062 inches

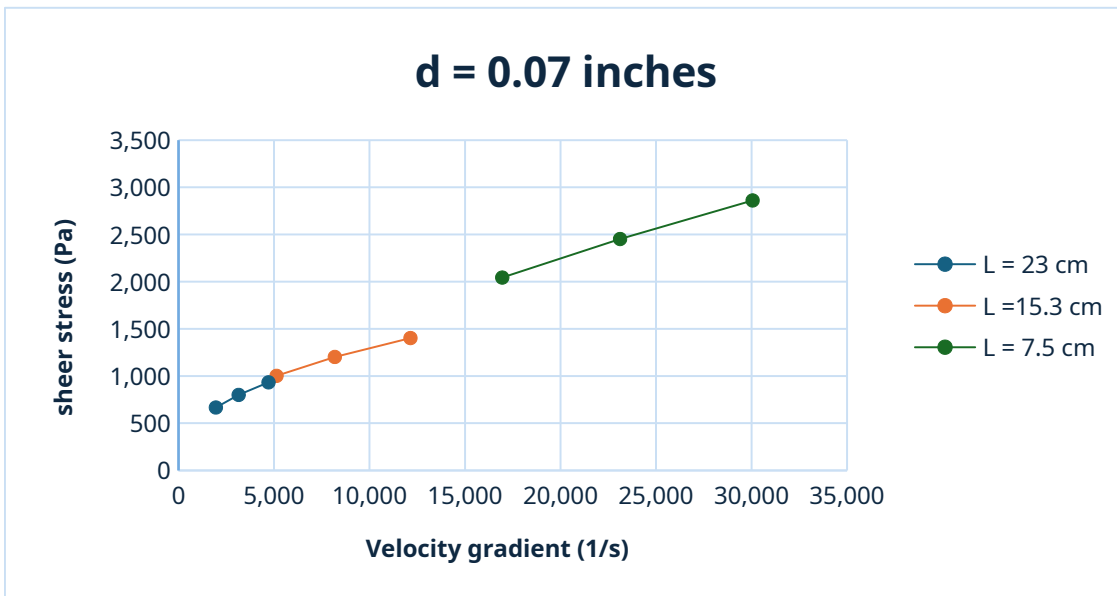


Figure 11: All ratios for diameter 0.07 inches

Figures 10 and 11 demonstrate fluid behavior at different pressures and lengths corresponding to the respective velocity gradients.

5. Discussion (15 pts)

The neutralization of the Carbopol solution was achieved using a solution of NaOH, with a pH range of 4-7 being the optimal pH. The final solution created had a pH of around 5-6 pH. [2]

The various plots were made for the different lengths and diameters of the tested capillary tubes. Lengths varying from 23 in, 15.3 in, and 7.5 in, and diameters 0.062 cm to 0.07 cm. For example, the respective m and n values for a length of 15.3 cm and a diameter of 0.062 in a yield of 43.47 Pa*s for m and .418 dimensionless for n . Additionally, the different L/D ratios yield different m and n values for each respective ratio. This can be understood by the nature of the Carbopol solution being a non-Newtonian fluid, which means that shear stress is directly proportional to the shear rate at a constant temperature and pressure. With the results shown, a linear graph is observed, indicating that as pressure increases, the flow rate also increases.

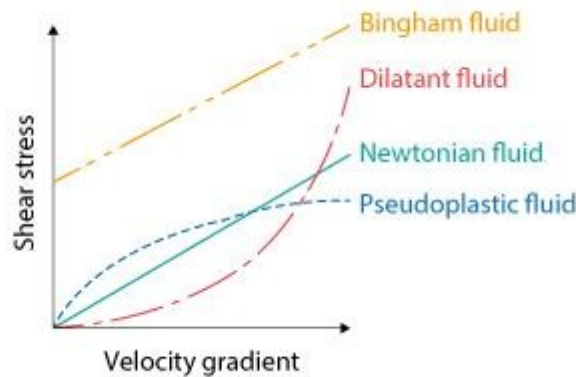


Figure 12: Plots of different fluid behavior [1]

The most effective way to compare the capillary tubes is to use the L/D ratio of each capillary tube. The increase in the value of the ratio L/D is observed, wherein the value of m increases, while the value of n decreases. This confirms that with the reduction in the value of the L/D ratio, the fluid exhibits pseudoplastic behavior, as indicated by the value of n , which approaches one, as shown in Equation 14. The values obtained are consistent with theory.

As shown in Figures 10 and 11, pseudoplastic behavior is evident, verifying the lab's goal and illustrating the shear thinning exhibited by the solution, as well as the non-Newtonian behavior of the 1 wt% Carbopol solution. As literature explains that a concentration above the .45 wt% threshold, where Carbopol will demonstrate non-Newtonian behavior [1]

When comparing the plots made from the experimental data collected, as the length of the capillary tube increases, the shear stress and velocity gradient also increase. When comparing the plots to the literature, a similar trend appears. So, the results obtained in the lab do agree with existing literature on the matter of “*Shear Stress V Velocity Gradient.*”

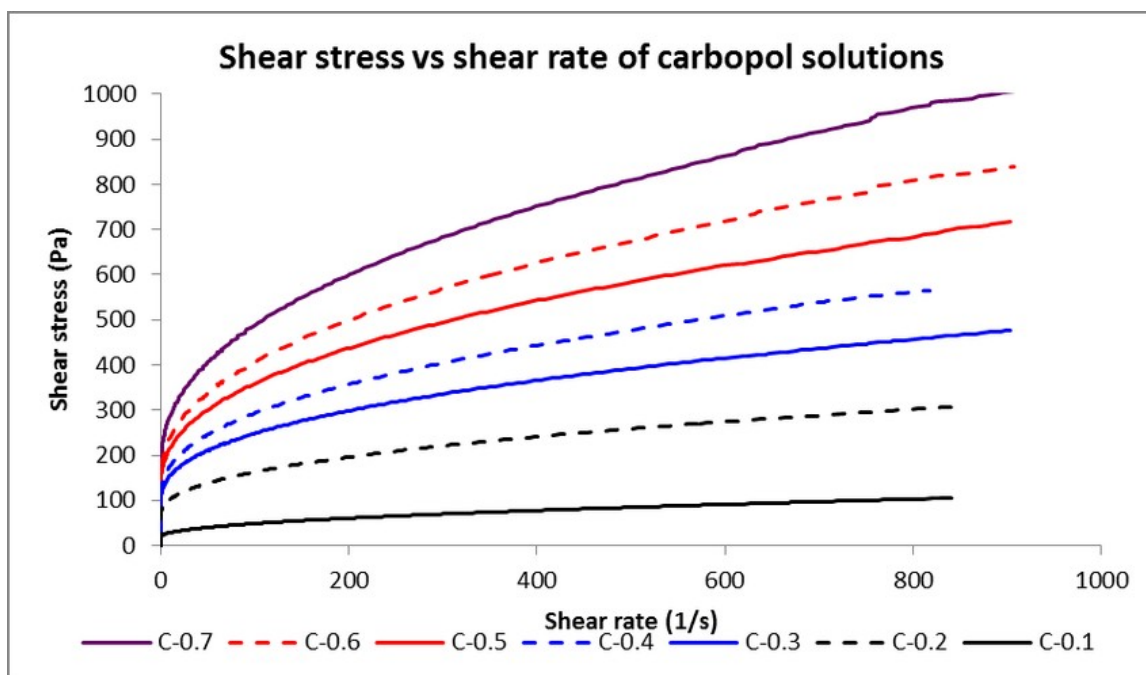


Figure 13: Literature plot of shear stress V velocity gradient of Carbopol Solution

There are also some assumptions that took place throughout the experiment, for example, the density was assumed to be constant, air bubbles did not affect the volume of the solution inside the cylinder, the rate of flow was constant, the pressure was constant, but once the rate of flow was obtained, the pressure gauge would gradually change slightly. All possible errors resulting from the assumptions are presented in the error analysis section. However, these assumptions are vital for achieving the desired result, as the required parameters cannot be obtained without them.

6. References (10 pts)

[1] Di Giuseppe, E., Corbi, F., Funicello, F., Massmeyer, A., Santimano, T. N., Rosenau, M., & Davaille, A. (2015). Characterization of Carbopol® hydrogel rheology for experimental tectonics and geodynamics. *Tectonophysics*, 642, 29–45. <https://doi.org/10.1016/j.tecto.2014.12.0057>.

[2] Department of Chemical Engineering, Viscosity of a Non-Newtonian Fluid Lab Manual, University of Illinois at Chicago, 2025.

[3] R.B. Bird, W.E. Stewart, and E.N. Lightfoot, Transport Phenomena, 2nd ed. John Wiley and Sons, Inc., 2002.

[4] NBCHAO. (n.d.). Working Principle, Classification and Application of Capillary Viscometer. NBCHAO. <https://en.nbchao.com/k/14058/>

[5] Felder, R.M., R.W. Rousseau, and L.G. Bullard, Elementary principles of chemical processes. 2020: John Wiley & Sons

[6] Perry, R. and D. Green, Perry's Chemical Engineers' Handbook 9th Edition. sl. 2018, McGraw-Hill.

Appendix I: Data Tabulation/Graphs (10 pts)

Table Experimental Data

run #	time (s)	pressure drop (pisg)	Weight (g)	length(c m)	diameter(inches)	Volume (mL)
1	11.61	50	6.7	23	0.062	6.6
2	10.41	60	7.8	23	0.062	7.7
3	12.58	70	12.5	23	0.062	12.4
4	12.85	50	13.6	15.3	0.062	13.4
5	11.2	60	19.0	15.3	0.062	18.8
6	11.15	70	26.3	15.3	0.062	25.9
7	12.36	50	55.9	7.5	0.062	55.2
8	11.56	60	62.7	7.5	0.062	61.9
9	11.28	70	88.7	7.5	0.062	87.6
10	10.9	50	8.2	23	0.07	8.2
11	11.56	60	14.5	23	0.07	14.3
12	10.73	70	19.5	23	0.07	19.3
13	9.65	50	20.2	15.3	0.07	19.9
14	9.43	60	30	15.3	0.07	29.6
15	9.41	70	46.8	15.3	0.07	46.0
16	7.93	50	70.1	7.5	0.07	69.2
17	9.51	60	92.9	7.5	0.07	91.7
18	8.06	70	117.7	7.5	0.07	116.2

Table Volume, volumetric flow rate, n, and m calculations

Volume (m ³)	Q (m ³ /s)	ln(Q/pi*R ³)	ln(delta P*R/L)	L/D ratio	n	m
6.6E-06	5.74E-07	5.92	6.38	146.05	0.52	12.13

Viscosity of Newtonian Fluids

University of Illinois

7.7E-06	7.43E-07	6.14	6.56	146.05	0.52	12.13
1.2E-05	9.87E-07	6.47	6.71	146.05	0.52	12.13
1.34E-05	1.04E-06	6.52	6.78	97.15	0.41	28.53
1.88E-05	1.68E-06	7.00	6.97	97.15	0.41	28.53
2.59E-05	2.33E-06	7.32	7.12	97.15	0.41	28.53
5.52E-05	4.47E-06	7.97	7.50	47.62	0.57	7.47
6.19E-05	5.35E-06	8.15	7.68	47.62	0.57	7.47
8.76E-05	7.77E-06	8.53	7.83	47.62	0.57	7.47
8.18E-06	7.51E-07	5.82	6.50	129.35	0.38	36.77
1.43E-05	1.24E-06	6.33	6.68	129.35	0.38	36.77
1.93E-05	1.80E-06	6.70	6.83	129.35	0.38	36.77
1.99E-05	2.06E-06	6.84	6.90	86.05	0.39	35.60
2.96E-05	3.14E-06	7.26	7.09	86.05	0.39	35.60
4.60E-05	4.88E-06	7.70	7.24	86.05	0.39	35.60
6.92E-05	8.72E-06	8.28	7.62	42.18	0.58	6.70
9.17E-05	9.64E-06	8.38	7.80	42.18	0.58	6.70
0.00011	1.44E-05	8.78	7.95	42.18	0.58	6.70

Table Shear stress, velocity gradients

sheer stress (Pa)	velocity gradient (1/s)	Sheer Stress (Pa) (no power law model)	velocity gradient (s ⁻¹) (no power law model)
590.10	1650.05	590.10	374.41
708.12	2336.26	708.12	485.00
826.14	3134.80	826.14	643.99
887.08	3723.37	887.08	683.08
1064.49	5759.24	1064.49	1097.66
1241.91	8327.70	1241.91	1520.16
1809.64	13051.11	1809.64	2914.63
2171.57	17878.54	2171.57	3494.60
2533.50	23329.05	2533.50	5069.15
666.24	1953.55	666.24	340.32
799.49	3147.33	799.49	562.89
932.74	4710.41	932.74	816.53
1001.54	5131.07	1001.54	936.66
1201.85	8183.18	1201.85	1426.86
1402.15	12142.74	1402.15	2214.98
2043.15	16947.89	2043.14	3954.41
2451.77	23117.35	2451.77	4371.16
2860.40	30055.84	2860.40	6537.13

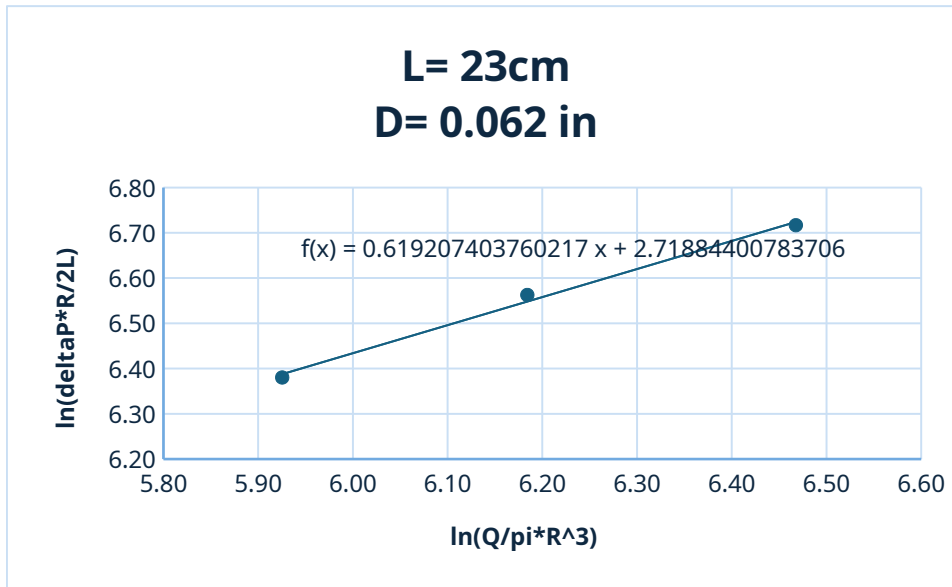


Figure 14: Relationship of pressure gradient and volumetric flow rate with length of 23 cm and diameter of 0.062 inches

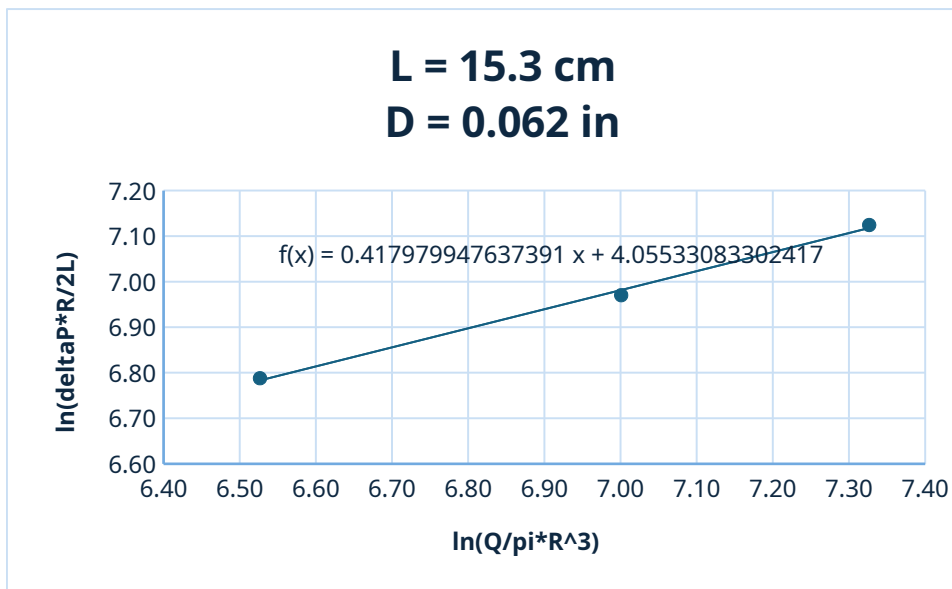


Figure 15: Relationship of pressure gradient and volumetric flow rate with length 15.3cm and diameter 0.062 inches

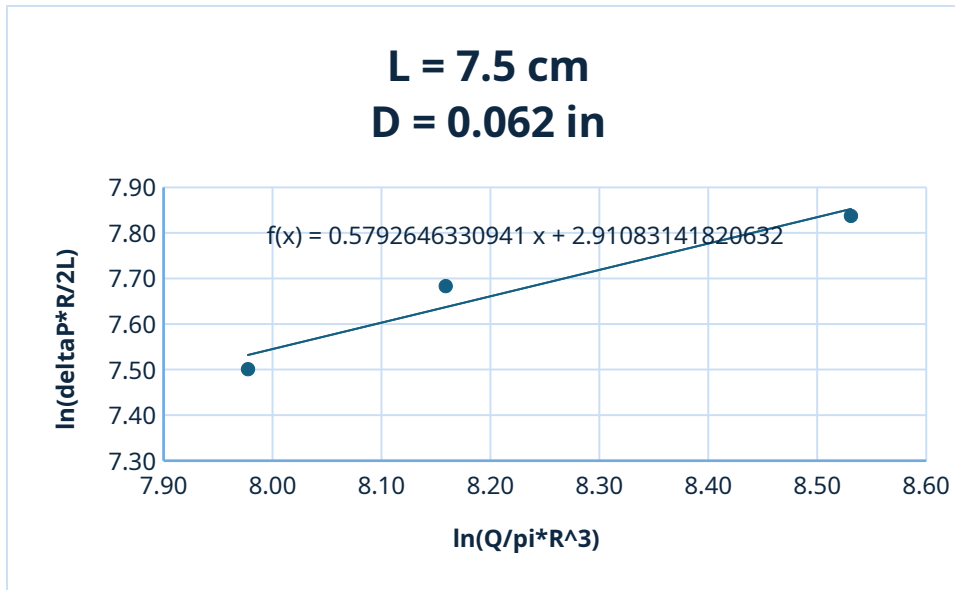


Figure 16: Relationship of pressure gradient and volumetric flow rate with length 7.5cm and diameter 0.062 inches

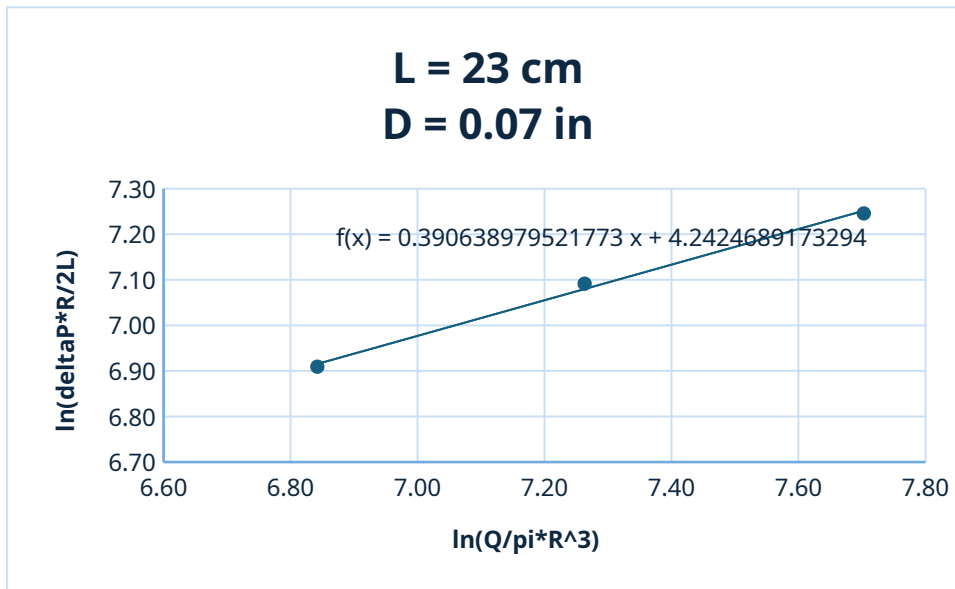


Figure 17: Relationship of pressure gradient and volumetric flow rate with 23 cm and diameter 0.07 inches

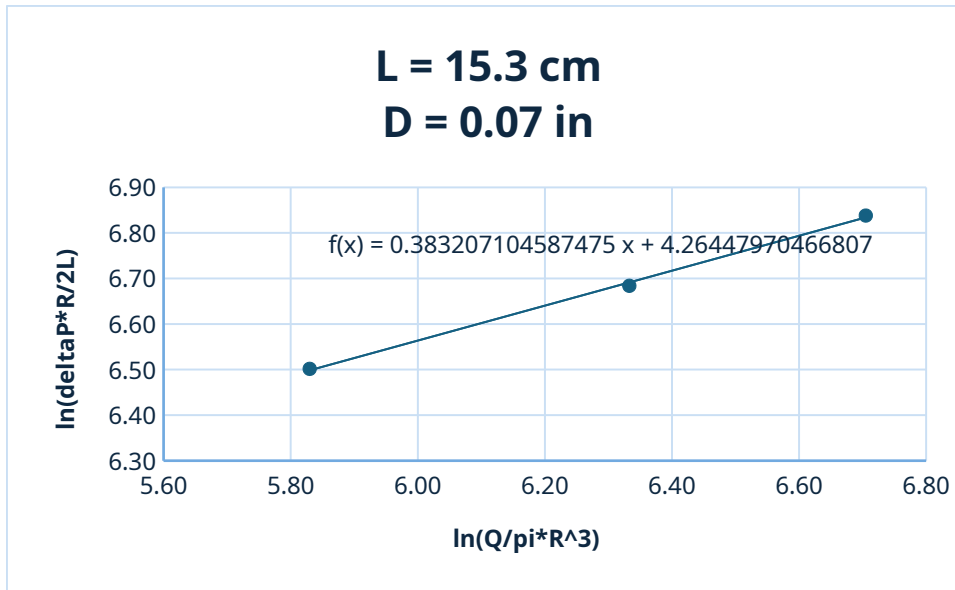


Figure 18: Relationship of pressure gradient and volumetric flow rate with length 15.3 cm and diameter 0.07 inches

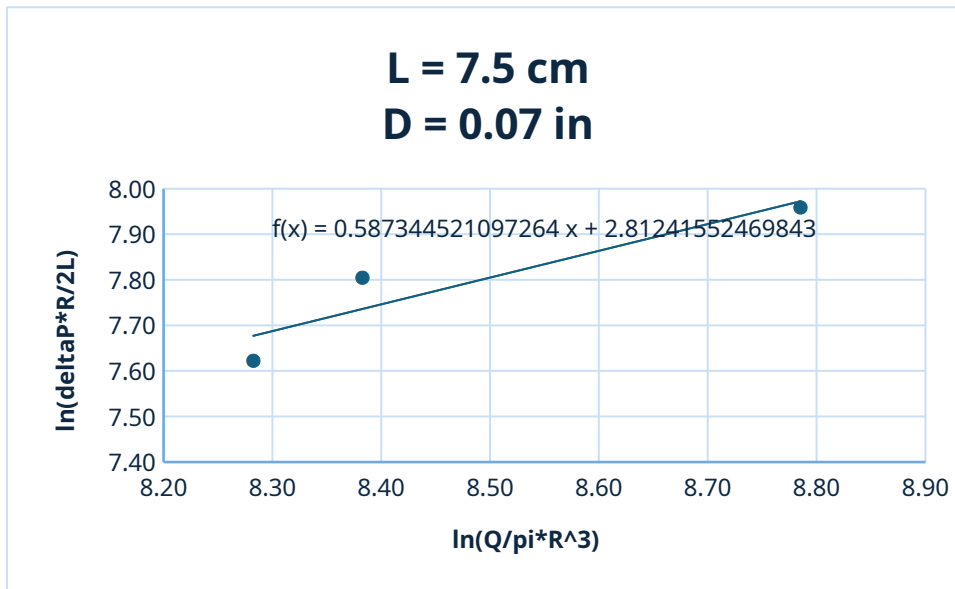


Figure 19: Relationship of pressure gradient and volumetric flow rate with length 7.5 cm and diameter 0.07 inches

Table : *N* parameter corresponding to Diameter and length

n	Diameter	
	(in)	
length (cm)	0.062	0.07
23	0.6192	0.3906
15.3	0.418	0.3823
7.5	0.5793	0.5873

8. Appendix II: Error Analysis (10 pts)

STD	$\ln(Q/\pi R^3)$	$\ln(Q/\pi R^3)$	STD	velocity gradient (s^{-1})	velocity gradient (s^{-1})
0.3640040					180.793
82	5.197354	5.92536223	96.81039673	374.4140353	24
0.2243820					309.635
09	5.735394	6.18415847	87.68478199	485.0046477	08
0.1351258					491.486
15	6.197435	6.46768652	76.252717	643.9921407	71
0.0954920					564.325
43	6.335632	6.52661596	59.37842132	683.0827142	87
0.0816815					932.225
49	6.837574	7.00093738	82.7182639	1097.661604	08
0.0455011					1387.93
08	7.235574	7.32657613	66.11538333	1520.167995	72
0.0178885					3020.80
83	8.013278	7.97750076	53.08264716	2914.637569	29
0.1127243					4378.33
38	8.384424	8.15897546	441.8660466	3494.604411	65
0.1088404					6301.92
06	8.74861	8.53092931	616.3843216	5069.154448	31
0.2868623					191.745
04	5.256167	5.82989113	74.28829804	340.3216266	03
0.1097767					451.873
06	6.113403	6.33295656	55.47212349	562.818142	9
0.0883337					684.303
12	6.528402	6.70506903	66.11541974	816.534391	55
0.5481382					312.952
21	5.746052	6.84232864	311.8575081	936.6677593	74
0.1986435					959.055
8	6.865949	7.26323585	233.9054736	1426.866202	25
0.1643755					1594.39
55	7.374251	7.70300226	310.2952859	2214.987994	74
0.0177320					3816.63
06	8.247124	8.28258836	68.89094541	3954.416609	47

0.1465098						5859.40
75	8.675803	8.38278372	744.1228291	4371.160093		58
0.0773005						7630.09
72	8.939855	8.78525413	546.479327	6537.134035		27

A standard deviation has been calculated on the duplicate set of velocity gradient calculations, and the calculation via the equation $\ln\left(\frac{Q}{\pi R^3}\right)$ to determine how much data and calculations deviated from each other.

ΔP values were set at the start of every test, except in prolonged experiments, where ΔP slightly decreased. This value has not been considered. There is a degree of uncertainty in our ΔP values, particularly in the measurement of the consistency index (m values). The value of the flow behavior index, n , has relatively higher accuracy, as it is not reliant on pressures.

In this experiment, the fluid density was assumed to be constant; however, small variations due to temperature, shear, and pressure will be introduced, resulting in small systematic errors in both volume calculations and rates. We have also assumed no wall slip; however, a smooth-walled tube would result in slip of Carbopol. If this is the case, the actual value of the shear stress would be smaller than calculated, making it appear more shear-thinning than it is.

Other sources of error could include human reaction time while operating the clock, which can act as a one- to two-second delay, and we disregarded the effects of gravity. Since the tubes weren't very long, disregarding gravity didn't matter too much, as the height wasn't very high.

9. Appendix III: Sample Calculations (10 pts)

To determine the density of the fluid, the mass and volume were measured and then divided.

$$\text{density} = \frac{\text{mass of gel}}{\text{volume of gel}} = \frac{324.7 \text{ g}}{320.7 \text{ mL}} = 1.01 \text{ g/mL}$$

The capillaries used were 23, 15.3, and 7.5 cm long, with diameters of 0.062 and 0.07 inches. A sample calculation for a 23 cm long and 0.062 in diameter capillary will be shown.

To calculate the flow rate, divide the grams of fluid collected by the time.

$$\text{flow rate} = \frac{\text{grams of fluid}}{\text{time}} = \frac{6.75 \text{ g}}{11.61 \text{ s}} = 0.581 \text{ g/s}$$

Then convert to mL per second

$$0.581 \frac{g}{s} \times \frac{1 mL}{1.01 g} = 0.580 \frac{mL}{s}$$

Now convert to flow rate

$$0.580 \frac{mL}{s} \times \frac{1 L}{1000 mL} \times \frac{1 m^3}{1000 L} = 5.8E-7 \frac{m^3}{s}$$

Next, the values were converted into SI units.

$$\begin{aligned} R &= 0.062 \text{ in} = 0.00079 \text{ m} \\ L &= 23 \text{ cm} = 0.23 \text{ m} \\ \Delta P &= 60 \text{ psia} = 413685.6 \text{ Pa} \end{aligned}$$

Next, the log for equation 13 can be calculated.

$$\begin{aligned} \log \frac{Q}{\pi R^3} &= \log \left(\frac{5.8E-7 \left(\frac{m^3}{s} \right)}{(3.14)(0.00079 \text{ m})^3} \right) = 2.57 \\ \log \frac{\Delta P}{2L} R &= \log \left(\frac{413685.6 \text{ Pa}}{2(0.0762 \text{ m})} (0.00079 \text{ m}) \right) = 3.33 \end{aligned}$$

Using excel functions “SLOPE()” and “INTERCEPT()” to find slope and y-intercept of the two $\log \frac{Q}{\pi R^3}$ vs. $\log \frac{\Delta P}{2L} R$

$$N = 0.5243$$

$$Y \text{ intercept} = 3.33$$

$$\begin{aligned} y_{\text{intercept}} &= \log \left\{ m \left[\frac{3n+1}{n} \right]^n \right\} \\ 10^{y_{\text{inter}}} &= m \left[\frac{3n+1}{n} \right]^n \\ m &= \frac{10^{1.71}}{\left[\frac{3(0.5243)+1}{0.5243} \right]^{0.47}} * \exp(3.33) = 12.1 \end{aligned}$$

These calculations are repeated for all trials.

The L/D ratios are then calculated.

$$\frac{L}{D} = \frac{.23\text{ m}}{2 * 0.0007874\text{ m}} = 146.05$$

Shear stress was calculated using the average m and n values.

$$\tau_{xy} = \pm m \left| \frac{dv_x}{dy} \right|^n$$

$$\tau_{xy} = 21.2 (0.95)^{0.4803} = 20.68\text{ Pa}$$

10. Appendix IV: Individual Team Contributions

NAME: Aaron Casas

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	6	In lab operating hours
Pre-Lab Editing	1	Finalized the safety portion of the lab
Experimental Objective (Prelab)	1	Formulated steps needed to be taken to complete the lab
Apparatus (prelab)	0	
Materials & Supplies (prelab)	0	
Procedure (prelab)	0	
Project Safety Evaluation (prelab)	1	Made safety guidelines
Final Lab Editing	3	Checked for completion of each section.
Executive Summary (final lab)	0	

Introduction (Final lab)	1	
Theory (Final Lab)	0	
Results (final lab)	1	
Discussion (final lab)	0	
References (final and/or prelab)	2	Referenced sources as well as in-text citations
Data Tabulation/Graphs (final)	2	Formulated the tables.
Error Analysis (final lab)	0	
Sample Calculations (final Lab)	2	Did sample calculations on excel.
Power Point Presentation	1	Added information to slides.
Total	21	

NAME: Hussein Rezk

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	8	Read and conducted an experiment
Pre-Lab Editing	3	Assisted in all parts
Experimental Objective (Prelab)	2	Assisted in objective creation
Apparatus (prelab)	1	Assisted in the assembly of the apparatus
Materials & Supplies (prelab)	1	Created a list of materials and supplies
Procedure (prelab)	2	Helped create a procedure for the experiment report
Project Safety Evaluation (prelab)	1	Made safety guidelines
Final Lab Editing	3	Made safety guidelines
Executive Summary (final lab)	2	Assisted in the formation of the summary
Introduction (Final lab)	0	
Theory (Final Lab)	3	Found valuable literature and theory to perform an

		analytical comparison
Results (final lab)	0	
Discussion (final lab)	4	Analyzed correlations, theory linearity, and the correctness of experiment results
References (final and/or prelab)	1	Tabulated references
Data Tabulation/Graphs (final)	0	
Error Analysis (final lab)	1	Conducted analysis on data
Sample Calculations (final Lab)	0	
Power Point Presentation	2	Created in collaboration with group
Total	33	

NAME: Adil Feratovic

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	8	Assisted in conduction of lab
Pre-Lab Editing	1	Made objective portion
Experimental Objective (Prelab)	1	Created the objective
Apparatus (prelab)	0	
Materials & Supplies (prelab)	0	
Procedure (prelab)	2	Created a procedure for the experiment report
Project Safety Evaluation (prelab)	0	
Final Lab Editing	4	Worked on formatting, sig figs
Executive Summary (final lab)	1	Worked on executive summary
Introduction (Final lab)	2	Worked on introduction
Theory (Final Lab)	2	Worked on theory
Results (final lab)	2	Worked on results
Discussion (final lab)	4	Worked on discussion
References (final and/or prelab)	0	Added references

Data Tabulation/Graphs (final)	0	Worked on calculations
Error Analysis (final lab)	0	Did an error analysis
Sample Calculations (final Lab)	2	Wrote out the sample calculations
Power Point Presentation	1	Worked on presentation
Total	32	

NAME: Anna Cai

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	8	Conducted experiment
Pre-Lab Editing	3	Edited grammar
Experimental Objective (Prelab)	0	
Apparatus (prelab)	2	Took photos of and wrote descriptions for the apparatus
Materials & Supplies (prelab)	1	Took into account materials and supplies needed
Procedure (prelab)	0	
Project Safety Evaluation (prelab)	0	
Final Lab Editing	3	Edited and finalized grammar
Executive Summary (final lab)	1	Assisted in writing the executive summary
Introduction (Final lab)	0	
Theory (Final Lab)		

Results (final lab)	2	Helped calculate results
Discussion (final lab)	2	
References (final and/or prelab)	0	
Data Tabulation/Graphs (final)	0	
Error Analysis (final lab)	0	
Sample Calculations (final Lab)	2	Wrote out sample calculations
Power Point Presentation	1	Finalized powerpoints
Total	25	